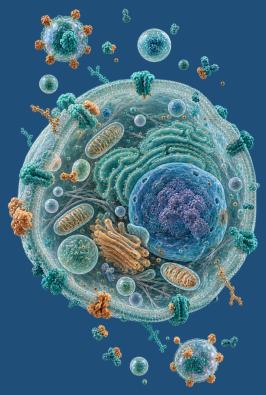
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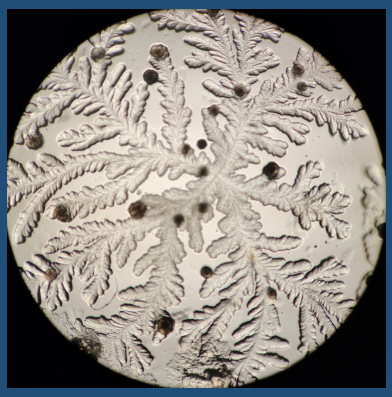
COURSE 77972

Physical Models of Biological Systems

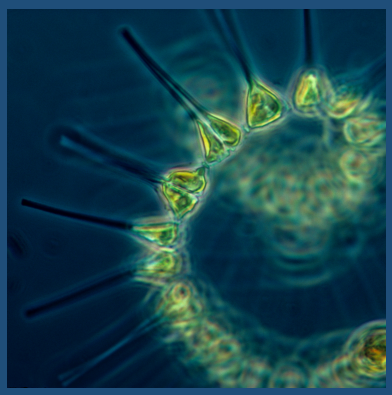
Lecture summaries • Exam revision



Cellular mechanisms



Microbial populations



Phytoplankton

2026–2027 | Semester A

Lecturer Dr. Amir Erez

Notes prepared by Eran Rehani

Cover: AI-generated conceptual cell illustration; branching colony: Korinna (2020), CC BY 4.0; phytoplankton: NOAA MESA Project, NOAA Photo Library, public domain, cropped.

Personal revision notes based on lectures and course materials.

0 Introduction

From Lecture 1 | 13 October 2026

Physical modelling connects biological mechanisms to observations. A useful model makes a question precise enough to calculate and test, with explicit assumptions and limitations.

0.1 Why living systems need selective descriptions

Living systems interact, adapt, and alter their surroundings: photosynthesis changed the atmosphere and hence the conditions for later life.

They maintain organization by exchanging matter and energy with their surroundings. This respects the second law: a steady biological state need not be thermodynamic equilibrium.

0.2 What a mathematical model contributes

A model specifies **variables**, **processes**, and their **relations**, then derives consequences not already inserted by hand.

- **Consistency:** equations make a verbal explanation precise and resolve competing effects quantitatively.
- **Prediction:** unexpected observations expose missing processes or faulty assumptions; precise biological forecasts remain difficult.
- **Inference:** observations constrain hidden rates or interactions when direct measurement is difficult or perturbs the living system.
- **Intuition:** a simple model isolates the dominant mechanism and helps us anticipate responses to change.

0.3 Choose the scale, then test the assumptions

Choose among molecules, cells, populations, communities, and ecosystems. Do space, fluctuations, or discrete individuals matter? Retain the relevant mechanism; add detail when it changes the conclusion.

Data and mechanisms complement each other. Ice-cream searches reveal annual periodicity without modelling its cause: predictive correlation need not explain a mechanism.

The modelling cycle: formulate → solve → compare with data → revise. Agreement supports usefulness, rather than proving every assumption. Chapter 1 applies this cycle to hidden viral turnover.

How to use these notes for revision

State assumptions, explain each equation term, and interpret results biologically. Attempt analytical homework before consulting solutions; these problems prepare you for the test. Vary simulation parameters to build intuition.

Further Reading

Amir Erez, course notes, Chapter 0, pp. 9–11; Philip Nelson, *Physical Models of Living Systems*, Chapter 2, "Physics and Biology."

1 Hidden HIV turnover

Lecture 1 | 13 October 2026

A constant viral load can conceal rapid production and removal. Blocking new infections exposes those processes: a two-variable model explains the initial shoulder and later decline, while showing what viral-load data cannot identify.

Revision goals

- Explain why a steady amount does not determine turnover.
- Translate biological assumptions into rate equations and solve them.
- Interpret the initial slope, equal-rate limit, late tail, and fitted rates.

1.1 Steady amounts and hidden turnover

The modelling principles in Chapter 0 lead to a concrete question: can a constant measured amount tell us how rapidly its contents are replaced?

Definition 1.1: Steady state and turnover

A steady state has constant macroscopic variables despite possible ongoing fluxes. For an amount W, input Q_{in}, and output Q_{out},

$$\dot{W} = Q_{\text{in}} - Q_{\text{out}}, \qquad \dot{W} = 0 \Rightarrow Q_{\text{in}} = Q_{\text{out}}. \tag{1.1}$$

The common flux can be large or small. Turnover is replacement of the contents while their amount can remain constant.

A biological steady state can consume energy and exchange matter; it need not be thermodynamic equilibrium.

Physical picture: the leaky container.

For liquid height h, cross-section A, density ρ, mass $m = \rho A h$, and gravitational acceleration g, force balance gives

$$p_{\text{bottom}} A = p_{\text{top}} A + mg, \tag{1.2}$$

$$\Delta p = \rho g h. \tag{1.3}$$

For a viscous Poiseuille outlet, $Q_{\text{out}} \propto \Delta p \propto h \propto W$. This assumes fixed cross-section and an appropriate flow regime; an arbitrary hole need not obey this law.

With constant input and $Q_{\text{out}} = \kappa W$ ($\kappa > 0$, units time^{-1}),

$$\dot{W} = Q_{\text{in}} - \kappa W, \qquad W^* = Q_{\text{in}}/\kappa. \tag{1.4}$$

Subtract the steady equation and integrate $\mathrm{d}(W - W^*)/(W - W^*) = -\kappa \mathrm{d}t$:

$$\boxed{W(t) = W^* + [W(0) - W^*]\mathrm{e}^{-\kappa t}.} \tag{1.5}$$

Deviations decay on time scale $1/\kappa$: higher volume increases removal and restores the level. This solution is exact for the linear outlet law. For example, $Q_{\text{in}} = 2\,\text{mL/min}$ and $\kappa = 0.1\,\text{min}^{-1}$ give $W^* = 20\,\text{mL}$ and relaxation time 10 min (added illustration).

1.2 The experiment behind the equations

Question. Untreated HIV can show an acute peak, a long approximately constant viral load, and a later rise. Does the middle stage mean slow replication? Alan Perelson's modelling and David Ho's treatment experiments challenge that interpretation: measured RNA load fell roughly a hundredfold over a week to ten days after protease inhibition. A perturbation reveals turnover that a steady measurement hides.

Minimum biological background.

Transcription copies DNA into messenger RNA; ribosomes translate its sequence into an amino-acid chain, which folds into a protein. HIV is a *retrovirus*: its RNA is reverse-transcribed into DNA, which can integrate into the host genome. Infected cells use that information to make viral components. Copying errors create variants; selection changes their relative abundances.

Ritonavir inhibits HIV protease, which cuts viral polyproteins into functional components. Already infected cells can still release RNA-containing particles, but impaired maturation prevents those new particles from infecting further cells in the ideal model. Mutations altering the protease's structure can reduce drug binding and confer resistance.

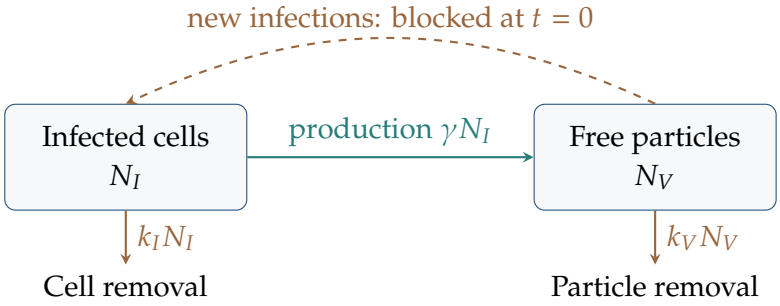


Figure 1.1: Read the model as fluxes: cells produce particles without being consumed. Clearance removes each population; treatment cuts the dashed infection feedback.

Combination therapy: the probability argument. If n distinct resistance mutations each have probability $p \ll 1$ per replication opportunity and are independent and required together, their joint probability is p^n. For M opportunities, the expected number is Mp^n. With $M = 10^9$, $p = 10^{-5}$, $Mp = 10^4$ but $Mp^3 = 10^{-6}$ (added toy example). This simultaneous-mutation model omits sequential evolution, selection, and cross-resistance; it illustrates independent barriers rather than determining a regimen.

1.3 Build and solve the model

1.3.1 Variables and assumptions

Time $t = 0$ marks treatment. These variables denote expected populations; large populations can approximate them with small relative fluctuations.

Symbol	Meaning and units
N_I, N_{I0}	Infected-cell count and initial value (cells).
N_V, N_{V0}	Free-particle count and initial value (virions).
k_I, k_V	Per-capita cell and particle clearance (time^{-1}).
γ	Production (virions cell^{-1} time^{-1}).
$\beta = \gamma N_{I0}$	Initial production (virions time^{-1}).

Here N_V includes released particles that cannot infect cells after treatment. Infectious and noninfectious particles are not distinguished. RNA copies per millilitre measure abundance, not infectivity: comparison requires a fixed conversion to the model observable. Concentrations may replace counts if all source terms are per unit volume.

Assume: steady viral load just before treatment; instantaneous, complete blocking of new infections; constant production per infected cell and clearance hazards; a well-mixed measured compartment. Reservoirs and cell heterogeneity are omitted. Pretreatment balance gives

$$\boxed{\beta = \gamma N_{I0} = k_V N_{V0}.} \tag{1.6}$$

This constrains the initial condition; production subsequently decreases.

1.3.2 From biological rules to differential equations

A cell's removal probability during Δt is $k_I \Delta t + o(\Delta t)$. Conditional on the current population, expected loss is $k_I N_I \Delta t$ to leading order. Taking expectations and $\Delta t \to 0$ gives the first equation below. Individual realizations fluctuate; their mean obeys the linear ODE exactly.

$$\dot{N}_I = -k_I N_I, \tag{1.7}$$

$$\int_{N_{I0}}^{N_I(t)} \frac{\mathrm{d}n}{n} = -k_I \int_0^t \mathrm{d}s, \qquad \ln \frac{N_I(t)}{N_{I0}} = -k_I t. \tag{1.8}$$

Exponentiating gives

$$\boxed{N_I(t) = N_{I0} \mathrm{e}^{-k_I t}.} \tag{1.9}$$

Constant per-cell hazard gives exponential survival of the source. Production is $\gamma N_I = \beta \mathrm{e}^{-k_I t}$, so

$$\boxed{\dot{N}_V + k_V N_V = \beta \mathrm{e}^{-k_I t}, \qquad N_V(0) = N_{V0}.} \tag{1.10}$$

Production declines as infected cells disappear; it does not switch off immediately.

1.3.3 Integrating factor: the derivation to reproduce

For $\dot{y} + a(t)y = f(t)$: multiply by $\mu = e^{\int a(t)dt}$, recognize $(\mu y)'$, and integrate from the initial time. Here $\mu = e^{k_V t}$:

$$\frac{d}{dt}(e^{k_V t} N_V) = \beta e^{(k_V - k_I)t}, \tag{1.11}$$

$$e^{k_V t} N_V(t) - N_{V0} = \beta \int_0^t e^{(k_V - k_I)s} ds$$

$$= \frac{\beta}{k_V - k_I}[e^{(k_V - k_I)t} - 1]. \tag{1.12}$$

Multiply by $e^{-k_V t}$; for $k_V \neq k_I$,

$$\boxed{N_V = N_{V0} e^{-k_V t} + \frac{\beta}{k_V - k_I}(e^{-k_I t} - e^{-k_V t}).} \tag{1.13}$$

The two modes combine particle clearance and loss of the source. Insert $\beta = k_V N_{V0}$:

Proposition 1.2: Viral load from an initially steady state

$$\frac{N_V}{N_{V0}} = \frac{k_V e^{-k_I t} - k_I e^{-k_V t}}{k_V - k_I}. \tag{1.14}$$

Two clearance time scales shape one observed trajectory.

1.4 Read the curve correctly

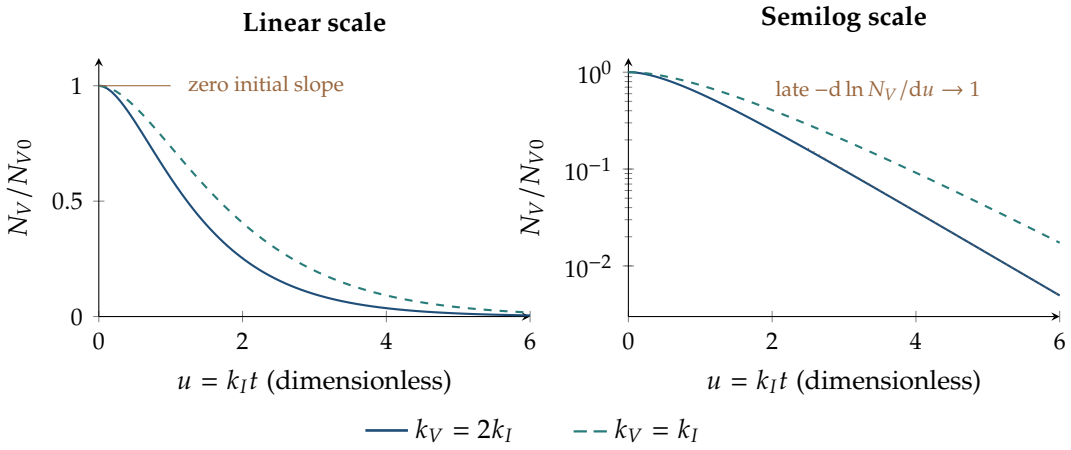


Figure 1.2: Model predictions, not experimental data. Both cases start horizontally. The solid curve approaches $2e^{-u}$ (dotted guide), while the dashed equal-rate curve retains a polynomial prefactor. On a semilog plot the late slope reveals the slower rate.

Initial shoulder.

Evaluate the ODE and its derivative using pretreatment balance:

$$\dot{N}_V(0) = \beta - k_V N_{V0} = 0, \qquad \ddot{N}_V(0) = -k_I \beta = -k_I k_V N_{V0} < 0. \tag{1.15}$$

The curve starts horizontally and immediately bends down; it is not flat over a finite interval. Its early expansion is

$$N_V/N_{V0} = 1 - \tfrac{1}{2} k_I k_V t^2 + O(t^3). \tag{1.16}$$

Curvature and the Taylor expansion supplement the lecture's shoulder argument.

Equal rates: remove the apparent singularity.

Set $k_I = k_V = k$ before integration: $(e^{kt}N_V)' = \beta$, hence $e^{kt}N_V - N_{V0} = \beta t$, giving

$$\boxed{N_V = (N_{V0} + \beta t)e^{-kt}.} \tag{1.17}$$

A polynomial prefactor replaces the distinct modes. For $\beta = kN_{V0}$,

$$N_V = N_{V0}(1 + kt)e^{-kt}, \qquad \dot{N}_V = -N_{V0}k^2te^{-kt}. \tag{1.18}$$

> **Remark**
>
> Equal rates do not remove the shoulder: the initial slope is still zero, consistent with official notes Eq. 1.9. A shoulder does not establish unequal rates.

Late decline and time scales.

If $k_V \gg k_I$, particles quickly track the source: $N_V \simeq (\beta/k_V)e^{-k_I t}$ after the fast transient. If $k_I \gg k_V$, the source disappears first and the tail decays with rate k_V. Thus the late logarithmic slope tends to $-\min(k_I, k_V)$, including the equal-rate case asymptotically. For a single exponential,

$$\tau = 1/k, \qquad t_{1/2} = \ln 2/k, \qquad r_{\text{app}} = \frac{\ln[N(t_1)/N(t_2)]}{t_2 - t_1}. \tag{1.19}$$

τ is mean lifetime under constant hazard; an interval's apparent rate r_{app} need not equal either mechanistic rate. A hundredfold fall in ten days gives $r_{\text{app}} = 0.461\,\text{day}^{-1}$ and apparent half-life 1.51 days (added calculation).

> **Example 1.3: Worked HIV calculation: two rates, one curve**
>
> Added problem: initially steady, $k_I = 1\,\text{day}^{-1}$, $k_V = 2\,\text{day}^{-1}$. Thus $\beta/N_{V0} = 2\,\text{day}^{-1}$. In Eq. (1.14), set $T = 1$ day:
>
> $$N_V/N_{V0} = 2e^{-t/T} - e^{-2t/T}, \tag{1.20}$$
> $$\dot{N}_V(0)/N_{V0} = -2/T + 2/T = 0, \tag{1.21}$$
> $$N_V(T)/N_{V0} = 2/e - 1/e^2 \simeq 0.600. \tag{1.22}$$
>
> Late: $N_V/N_{V0} \simeq 2e^{-t/T}$. The tail measures the slower rate; cancellation of slopes creates the shoulder. Exchanging rate labels preserves the curve.

Checks and common pitfalls.

ODE terms must have units population/time; the solution must return N_{V0} at $t = 0$; balance fixes the initial slope. A negative mode coefficient is allowed: the total is nonnegative, as the equivalent integral shows:

$$N_V = N_{V0}e^{-k_V t} + \int_0^t \beta e^{-k_I s}e^{-k_V(t-s)}ds \geq 0. \tag{1.23}$$

1.5 What can an exam answer conclude?

Definition 1.4: Structural identifiability

A parameter is structurally identifiable if ideal observations distinguish it from other choices. This differs from whether noisy, finite data constrain it accurately.

- **Count parameters:** measured N_{V0} leaves β, k_I, k_V; pretreatment balance leaves two rates.
- **Check ambiguity:** Eq. (1.14) is invariant under $k_I \leftrightarrow k_V$. Viral load alone can at best identify the unordered pair. Cell measurements or justified biological information must assign labels. Also $\beta = \gamma N_{I0}$ does not separate its two factors.
- **Validate:** count independent features, not raw points; nearby measurements may be correlated. More independent constraints than parameters permit a test, but do not guarantee identifiability. Overfitting means fitting noise with excess flexibility.
- **Respect scope:** rapid decline supports rapid hidden turnover. This model omits long-term disease progression and latent reservoirs. Dormant infected cells can retain viral DNA without producing particles, so undetectable blood RNA does not imply eradication. Reservoir activation was discussed as exploratory, not established by this calculation.

Self-check before the exam

1. Derive the integrating-factor solution, starting from production and loss.
2. Why are the initial slope zero and the initial curvature negative?
3. Obtain the equal-rate limit. Does the shoulder disappear?
4. Which process sets the late slope in each separated-scale limit?
5. Why can viral load alone not assign the two rate labels?

Core chain: steady amount → unknown fluxes → perturbation → time scales → qualified inference.

Further Reading

- Amir Erez, *Physical Models of Living Systems: Lecture Notes*, `models-notes-2026.pdf`: Chapter 0 and Chapter 1, §§1.1–1.7, printed pp. 9–18. Used to check conventions and results; later extensions are outside this summary.
- Philip Nelson, *Physical Models of Living Systems*, supplied Penn edition: Chapter 1, "Virus Dynamics," §§1.2.1–1.2.6 (biology, semilog plots, rate equations, fitting).
- A. S. Perelson et al., "HIV-1 dynamics in vivo", *Science* **271**, 1582–1586 (1996): original study, citation and abstract.